

PIERRE 24 S, SD, USA

WMO: 726630

Lat: 44.0194N

Lon: 100.3530W

Elev: 647

StdP: 93.79

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 94085

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -22.8	(c) -20.3	(d) -26.7	(e) 0.4	(f) -22.7	(g) -24.0	(h) 0.5	(i) -19.9	(j) 13.3	(k) -8.1	(l) 12.0	(m) -6.7	(n) 3.6	(o) N/A	(p) 0.549

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.5	35.3	21.1	33.0	20.8	31.0	20.2	23.8	30.8	22.7	29.4	21.7	28.3	3.8	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.8	17.8	27.1	20.6	16.5	26.3	19.6	15.5	25.4	74.6	31.0	70.3	29.7	66.4	28.1	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.6	9.2	7.9	-26.6	38.9	2.2	2.5	-28.2	40.7	-29.5	42.2	-30.7	43.6	-32.2	45.4
			-26.9	25.4	2.6	1.1	-28.8	26.2	-30.3	26.8	-31.8	27.5	-33.7	28.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.3	-6.2	-6.1	1.9	7.2	13.5	19.5	23.6	22.1	17.7	9.2	2.1	-5.4		
	DBStd	12.09	7.19	7.69	7.74	6.02	4.85	3.64	3.40	3.34	4.60	5.58	6.48	6.80		
	HDD10.0	2171	503	453	266	119	22	1	0	0	3	82	246	476		
	HDD18.3	4064	762	685	509	336	162	28	3	7	66	285	488	734		
	CDD10.0	1566	0	1	15	35	131	286	422	375	234	58	9	0		
	CDD18.3	418	0	0	1	1	13	64	166	123	46	3	0	0		
	CDH23.3	4563	0	0	13	42	177	651	1822	1248	545	64	1	0		
	CDH26.7	1937	0	0	2	9	50	221	878	547	209	21	0	0		
(14) Wind	WSAvg	3.5	4.2	3.8	4.0	4.1	3.7	2.8	2.3	2.8	3.2	3.7	3.6	3.7		
(15) Precipitation	PrecAvg	483	11	13	23	52	80	92	60	50	40	38	14	10		
	PrecMax	684	50	34	75	103	153	190	149	136	145	115	40	30		
	PrecMin	218	0	0	1	7	15	25	7	2	0	4	0	0		
	PrecStd	107	11	9	19	28	39	40	38	30	34	35	12	9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.8	17.9	25.4	27.7	31.2	34.6	39.9	37.2	35.7	29.7	22.2	13.8		
		MCWB	5.8	8.9	12.0	14.2	18.9	20.9	21.0	20.6	18.5	15.3	10.7	6.5		
	2%	DB	7.9	10.7	20.6	23.7	27.8	31.2	36.2	34.3	31.6	24.4	17.6	9.3		
		MCWB	3.3	5.2	9.6	12.7	17.5	21.1	21.7	20.8	18.6	13.6	9.0	3.6		
	5%	DB	5.3	7.1	16.9	20.4	25.0	29.2	33.7	31.9	28.9	21.0	14.3	5.9		
		MCWB	1.8	3.2	8.4	10.8	16.3	19.9	21.6	20.6	18.0	11.8	7.1	1.7		
	10%	DB	3.4	4.4	13.1	17.4	22.4	27.2	31.4	29.8	26.5	18.0	11.3	3.5		
		MCWB	0.8	1.4	7.0	9.0	14.8	19.3	20.8	19.9	16.8	10.4	5.5	0.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.2	9.1	13.0	15.1	20.6	24.0	25.7	24.4	22.1	16.7	11.6	6.9		
		MCDB	12.7	18.3	19.4	24.8	28.3	31.5	32.6	32.3	28.9	27.3	21.4	13.4		
	2%	WB	3.7	6.3	10.5	13.0	19.0	22.6	24.3	23.0	20.5	14.6	9.2	4.0		
		MCDB	7.9	12.5	20.0	22.1	25.7	29.0	30.8	29.9	27.5	23.0	17.2	9.3		
	5%	WB	2.1	3.7	8.8	11.3	17.4	21.5	23.1	21.9	19.3	12.9	7.3	2.2		
		MCDB	5.1	7.6	16.8	19.1	23.1	27.5	30.0	28.0	26.1	20.0	13.6	6.0		
	10%	WB	1.0	1.9	7.0	9.8	15.6	20.3	22.1	20.9	18.1	11.2	5.5	0.7		
		MCDB	3.6	4.9	13.5	16.2	20.9	26.2	29.2	27.0	24.6	17.7	10.6	3.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.9	10.2	11.7	12.4	12.0	12.0	13.5	13.1	13.5	11.8	11.2	9.2		
		MCDBR	12.8	13.9	18.0	18.7	15.8	15.1	16.7	16.4	17.1	17.5	15.8	12.7		
	5% WB	MCWBR	8.7	9.0	9.6	10.0	7.7	7.4	5.9	6.3	6.7	8.8	8.9	8.1		
		MCWBR	12.1	13.5	17.5	17.2	13.9	13.6	14.0	13.7	14.6	15.9	15.2	12.1		
(40) Clear-Sky Solar Irradiance	taub		0.258	0.267	0.289	0.332	0.352	0.365	0.383	0.378	0.342	0.303	0.269	0.256		
	taud		2.533	2.525	2.527	2.396	2.421	2.411	2.392	2.384	2.462	2.544	2.552	2.524		
	Ebn at Noon		895	944	959	932	918	903	882	874	885	886	872	861		
	Edh at Noon		65	79	90	112	113	115	116	113	96	77	64	60		
(44) All-Sky Solar Radiation	RadAvg		1.91	2.86	3.92	4.95	5.94	6.72	6.92	5.80	4.64	3.07	2.07	1.56		
	RadStd		0.15	0.23	0.31	0.36	0.56	0.45	0.27	0.42	0.31	0.36	0.20	0.10		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page